Zhiheng JIN

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Personal Website | LinkedIn

Research Interests

System Identification, Real-Time Optimization of Autonomous Systems, Multi-Agent Systems, Machine Learning, Control Theory

Education

Nanjing University of Aeronautics and Astronautics Sept 2024 – Present
Master of Research in Control Science and Engineering (GPA: 85/100)
Supervisor: Lecturer Dan Yu

Shenyang Aerospace University Sept 2019 – Feb 2021
Major in Mechanical Engineering, then Aircraft Design and Engineering

Shenyang Aerospace University Feb 2021 – Jun 2023
Bachelor of Engineering in Aircraft Design and Engineering (GPA: 82/100)

Publications

Under Review

- **Manipulator Motion Planning via Reactive Flow-Diffusion and Refinement**
  Zhiheng Jin, Dan Yu — 1st Author (Master)
  Submitted to PIMechE Part I: Journal of Systems and Control Engineering.

- **Disturbance Prediction-based Koopman Bilinear Model Predictive Control for Control-Affine Systems**
  Dan Yu, Hongyu Zhu, **Zhiheng Jin** — 3rd Author (Master)
  Submitted to IEEE Robotics and Automation Letters (RA-L), 2026.

Journal Publications

- **Humanity Enters the Age of Aviation**
  Zhiheng Jin, Xin Wen — 1st Author (Bachelor)
  Space Exploration (Popular Science), 2021.

- **Exploration of Centrifugal Impeller Manufacturing via Investment Casting**
  Yiqing Tao, Bing Yang, **Zhiheng Jin**, Xize Yu, M Chen, Y Shi, F Liu — Co-Author (Bachelor)
  China Foundry Equipment Technology, 2021.

- **Shortest Path Research in Games Based on Graph Theory**
  Zhiheng Jin, Zhejia Zhou, Junpeng Wang — 1st Author (Bachelor)
  China Science Investment, 2020.

Patents

- **Multi-Arm Trajectory Planning Method Based on Lightweight Diffusion and Reactive Flow Matching**
  Dan Yu (Supervisor), **Zhiheng Jin** — 2nd Author (Master)
  Patent Application, submitted in 2026.

- **Floating Strip Garbage Collection Device**
  Pengqun Xiao, **Zhiheng Jin**, Yicheng Shen, L Shao, Y Li, Jingyu Yang — 2nd Author (Bachelor)
  Utility Patent, 2022, Patent No: CN217460494U.

Undergraduate Thesis

- **Comparative Study of Population-Based Metaheuristic Algorithms**
  Zhiheng Jin, 2023. Supervised by Associate Prof. Yiming Du.

Research Projects

Intelligent Robotic Arm for a Digital Drilling and Workover Platform 2024

- Designed an intelligent manipulator system integrating visual servoing, cooperative transport, and compliant control.
- Verified design feasibility via simulation and authored technical reports and presentations.
- Awarded **National Third Prize**, "Qinchuangyuan–CPSM Cup" 11th China Graduate Energy Equipment Innovation Competition, Xi'an, Oct 2024.

Intelligent Inspection of Power Transmission Lines Using UAV Swarms 2024

- Designed and trained YOLO-based networks for visual defect detection and dataset augmentation.
- Integrated trained models into UAV platforms and participated in field tests and documentation.

Microfluidic-Based Viscosity Measurement 2021

- Proposed a system to measure fluid viscosity using microfluidic chips and mechanical propulsion.
- Designed a motor-driven propulsion system using a microcontroller.
- The project passed university evaluation and won **First Prize in the Liaoning Province Physics Competition (Bachelor Group)**.

Internship Experience

Jiangsu Industrial Technology Research Institute 2025 – Present

- Built control models based on experimental test data for ceramic heating plate temperature-control systems.
- Conducted finite element simulations to analyze thermal behavior and support controller design.
- Optimized the microchannel cooling design of ceramic plates to improve heat dissipation performance.
- Tuned PID parameters and evaluated temperature-control performance under experimental conditions.

Liaoning General Aviation Research Institute 2021

- Conducted model preprocessing, lightweight optimization, and drawing verification using SolidWorks and UG NX.
- Detected and corrected multiple model-drawing inconsistencies, reducing potential manufacturing risks.
- Participated in the "Composite Manufacturing Process of Irregular Aircraft Components" project, engaging in prepreg and liquid molding of carbon-fiber wing models and gaining full-cycle process experience.

Lab Experience

Acoustic Metamaterials and Wave Manipulation 2020

- Built simulation models using COMSOL Multiphysics, configured acoustic field parameters, and analyzed simulation results.

Skills

Programming: Python, C, XML.

Packages: PyTorch, MuJoCo, Matplotlib, NumPy, SciPy, OpenCV.

Software: MATLAB, LaTeX, SolidWorks, AutoCAD, CATIA.

Soft skills: Self-motivation, interdisciplinary experience, accountability, research leadership, verbal & written communication, teamwork, time management, critical thinking, curiosity.

Selected Courses

Mathematics Matrix Theory, Probability and Statistics, Advanced Statistics, Linear Algebra, Advanced Mathematics, Numerical Methods.

AI & CS: Intelligent Robotics & Applications, Internet of Things (IoT), Fundamentals of Computer Science, Computer Programming.

Theory: Optimal Control, Optimization Design, Automatic Control, System Identification, Computer Control.

Basic: Mechanism Principles, Theoretical Mechanics, Digital Circuits, Analog Circuits.

Honors and Awards

- **Outstanding Graduate Student Award** 2024 – 2025
- **National Encouragement Scholarship** 2020
- **University Comprehensive Academic Scholarship** 3 times
- **Outstanding Student Award** 2020
- **Japan Eurasia Foundation Scholarship** 2021

Teaching Experience

Automatic Control Theory Teaching Assistant

Introduction to Control Theory, Transfer Functions, Time-Domain Analysis, Root Locus Method, Frequency-Domain Analysis and Compensation, Discrete Control Systems, Nonlinear System Analysis, and Introductory Modern Control.

Other Experience

Member of the Drama Troupe 2019

- Performed in 2019 Freshman Gala.

Member of the Debate Club 2019 – 2020

- Participated in multiple debate competitions, winning first prizes.

Academic Style Development Center 2020

- Tutored Linear Algebra and Theoretical Mechanics, helping 20+ students pass exams.

Member of Tianyi Drama Club 2025

- Supported the original play "Wukong" at Jiangsu Grand Theatre, managing props and stage coordination.